Q.28 What are advantages of 3---phase system over 1-phase system.

Q.29 Differentiate between motor and generator.

Q.30 Describe the flux control method of speed control of d.c series motor.

Q.31 Write a short note on methods of measurement power in a 3-phase circuit.

Q.32 Briefly explain the effect of change in excitation of a synchronous motor with phasor diagram

Q.33 Explain the significance of back e.m.f. in a d.c. motor.

Q.34 Write down the various applications of synchronous motor.

Q.35 Draw the torque-slip curve of a 3-phase induction motor. also mark the stable and unstable operating region on it.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Define starter and make a list of methods of starting for 3-phase induction motor. Explain anyone method with neat diagram.

Q.37 Explain the construction, principle and working of a Stepper motor with neat sketch.

Q.38 Explain the construction, principle and working of an alternator with neat sketch

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4th Sem / Mechatronics

Subject:- DC and AC Machines

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 At synchronous speed, the slip of the induction motor will be

a) zero b) 0.5

c) 1 d) infinite

Q.2 Voltage and current relationship in a 3 phase Star connected System

a) $V_L = \sqrt{3}Vph, I_L = Iph$ b) $V_L = Vph, I_L = \sqrt{3}Iph$

c) $V_L = Vph, I_L = Iph$ d) $V_L = 3Vph, I_L = 3Iph$

Q.3 Brushes of DC Machines is made up of_______

a) silicon steel b) Brass

c) Cast iron d) Carbon

Q.4 For ceiling fans, generally the single phase induction motor used is

a) Shaded pole

b) Capacitor start

c) permanent capacitor type

d) Capacitor start and capacitor run

Q.5 The shaft of an induction motor is made up of

a) silicon steel b) mild steel
c) brass d) cast iron

Q.6 The frequency of voltage generated in large alternators in India is
a) 0Hz b) 25Hz
c) 60Hz d) 50Hz

Q.7 For 3 stack 16 tooth rotor VR stepper motor, the stepping angle is______°.
a) 7.5 b) 15
c) 30 d) 90

Q.8 Maximum speed of a synchronous machine for 50 Hz supply frequency is______ R.P.M
a) 1500 b) 3000
c) 20000 d) 30000

Q.9 The motor which is used in the control system are called
a) Stepper motor
b) Linear induction motor
c) Servo motor d) Synchronous motor

Q.10 _____ winding is used for self-starting of synchronous motor.
a) Field b) Armature
c) damper d) none of the above

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 The direction of 1-phase induction motor can be reversed by_______.

Q.12 Write any one application of DC series motor.

Q.13 Over excited synchronous motor working at no load behaves like a ______

Q.14 Give expression of synchronous speed Ns=

Q.15 Give any two applications of slip ring induction.

Q.16 Universal motor can work on____ and ____ supply.

Q.17 The rating of alternators is usually expressed in_____.

Q.18 The yoke of DC machine are made up of______.

Q.19 A 4-pole wave wound d.c. motor will have________ parallel paths.

Q.20 E.M.F equation of DC motor E_b=

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Explain the working principle of a 3-phase induction motor.

Q.22 Draw and explain the torque vs speed characteristics of a d.c shunt motor.

Q.23 Write a short note on universal motor and its application.

Q.24 Explain how a synchronous motor is made the self-starting.

Q.25 Explain in brief the various methods of speed control of 3-phase Induction motor.

Q.26 Make a list of types of d.c. motots.

Q.27 Explain what you would understand by armature reaction in a d.c. generator and its effects.